

Phrasal Provenance: An LLM-Assisted Methodology for Elite Network Research

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Abstract

We present a methodology for LLM-assisted research that anchors every claim to a specific sentence in a source document. Using European institutional elites as a case study—commissioners, DGs, and MEP leaders across seven commission terms (1995–2029)—we demonstrate a pipeline that scrapes from controlled sources, normalises data into a relational database with mandatory provenance links, uses LLMs for phrase-level fact extraction with JSON manifests, and cross-validates every claim against source text to detect hallucinations. The system surfaces substantive findings including the collapse of the French *grande école* pipeline into the Commission and a stark asymmetry in elite network ties between commissioners (12 shared networks) and MEP leaders (0 exclusive networks). We report on failure modes encountered—LLM hallucination and entity duplication from non-deterministic prompt outputs, and the “CFR is not ECFR” problem in string-similarity deduplication—and describe the methodological choices that address each.

1 Introduction

The study of elite networks has historically proceeded through detailed examination of who occupies key institutional positions [5]. For decades this research relied on manual methods—biographical analysis, archival work, surveys of officeholders—with limited scalability [3]. Large language models offer the possibility of scaling this work: extracting structured facts from unstructured biographies, identifying patterns across hundreds of careers, and surfacing connections that would take years of manual work to discover. But LLMs hallucinate—they generate plausible-sounding but factually incorrect claims [1]. In research, a claim without a traceable source is not a finding.

Crucially, hallucination is primarily a property of *generation*—the task of producing novel text from a model’s internal distribution. The task we employ the LLM for is fundamentally different: *verification* of whether a specific sentence from a source document supports a specific factual claim. This is a form of natural language inference, not generation—the model is not asked to invent, but to judge. Recent work on fine-grained citation grounding demonstrates that LLMs can reliably anchor claims to specific source sentences when constrained to do so [6]. By numbering phrases and requiring the LLM to cite a specific phrase index for each extracted fact, we move from generation (high-hallucination regime) to citation (low-hallucination regime).

This paper describes a methodology that treats the LLM as an extraction engine, not an authority. Every fact in our database is linked to a specific sentence in a specific document via a phrase index. The LLM’s output is a hypothesis, verified against the source before acceptance. The system currently holds 2,600+ facts across 550+ entities with sentence-level provenance,

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supporting analysis of seven European Commissions from Santer (1995) through Von der Leyen II (2024–2029).

2 The Hallucination Problem

LLMs hallucinate because they generate text by sampling from a learned probability distribution over tokens. When the model lacks grounding in external facts, this distribution can produce plausible-sounding but incorrect statements [1]. Manakul et al. (2023) demonstrate that hallucinated outputs are detectable through self-consistency: when an LLM is asked the same question multiple times, hallucinated answers tend to diverge and contradict each other, while factual answers remain consistent across samples [4]. This reveals that hallucination is a property of *unconstrained* generation: the model’s internal distribution over tokens is underdetermined by the prompt, and stochastic variance produces divergent outputs.

Crucially, Kadavath et al. (2022) show that LLMs are substantially better at *evaluating* their own outputs than at *generating* correct ones—they can detect errors in existing text at much higher rates than they can avoid producing errors during generation [2]. This generation-verification gap is the foundation of our approach: we do not ask the LLM to invent facts. We ask it to cite a numbered phrase from a source document that supports a factual claim. The task changes from “produce text from your internal distribution” to “does this specific sentence support this specific claim?”—a form of natural language inference with dramatically lower hallucination rates.

LongCite (2024) provides direct evidence that LLMs can reliably anchor claims to specific source sentences when trained and prompted to do so [6]. Our methodology extends this insight: by numbering phrases and requiring the LLM to return a phrase index for each extracted fact, we build a system in which every claim in the database is traceable to a specific sentence in a source document. The LLM acts not as a knowledge base but as a structured extraction engine with mandatory provenance.

3 System Architecture

The pipeline has five stages: scrape, extract, verify, resolve, and render.

3.1 Database Schema

The database implements a graph-style ontology on a standard relational backend. Every piece of data is stored as a triple: an *entity* (person, organisation, institution, commission, or source document) connected to a *property value* by a *predicate* (`educated_at`, `affiliated_with`, `served_on_commission`, `held_portfolio`, `from_country`). The property value may be another entity (forming a graph edge) or a literal value (a date, a degree, a country name).

The predicates are drawn from the substantive concerns of elite research: education (`educated_at`, `studied_field`, `held_degree`) to track elite formation; organisational ties (`affiliated_with`, `member_of`) to map network membership; career trajectories (`served_on_commission`, `held_portfolio`, `from_country`); and institutional position (`served_on_ep`). Each fact—each triple—carries mandatory provenance: the source document from which it was extracted, the exact sentence (stored as `quote_text`), and its position within that document (`phrase_index`). This transforms the database from a collection of assertions into an auditable knowledge graph: every edge can be traced to a specific sentence in a specific source. The four core tables are:

- **entity**: All nodes in the graph—persons, organisations, educational institutions, commissions, and source documents—with a type, category, and optional country code.

- **fact**: Directed edges connecting entities (`educated_at` links a person to a university) or attaching literal properties (`held_degree` links a person to “PhD”). Includes confidence and qualifier fields.
- **citation**: Metadata for each source document—URL, access date, source type (Wikipedia, Wikidata, Commission CV PDF, etc.).
- **provenance**: Every fact’s audit trail—the `citation_id`, the `phrase_index` (sentence number), and the exact `quote_text` that supports the claim.

Facts without provenance are deleted rather than retained. This is not a cleanup operation but a structural invariant: the knowledge graph is self-auditing.

3.2 Phrase-Level Extraction

The core innovation is phrase-indexed extraction. Rather than sending an article to an LLM and accepting unstructured output, we:

1. Split source text into numbered sentences (50–80 per document).
2. Present the LLM with numbered phrases and a structured prompt: “Extract every organisation that X has been a member of. For each: phrase number, organisation name, role, reasoning.”
3. Parse the JSON response into a *manifest*—a durable intermediate artifact stored as JSON.
4. Cross-check: retrieve the sentence at the claimed phrase index and verify it contains the asserted information.

The manifest preserves the LLM’s full output including reasoning and phrase indices, making extraction auditable and reproducible.

4 Failure Modes

We encountered three categories of failure:

4.1 Hallucination (18% of extractions)

In our commissioner education extraction run, 18% of LLM outputs were unsupported by the source text at the claimed phrase index. Typical case: the LLM inferred “Harvard” from context (“studied at an Ivy League university”) when Harvard was not explicitly named. Phrase-index cross-checking caught these.

4.2 Non-Deterministic Extraction

Running identical prompts on the same source text produced different sets of extracted organisations on different runs. The LLM’s non-determinism meant that manifest regeneration changed the database. This is tolerable for additive processes (new facts are added, existing ones persist) but problematic for analytical reproducibility. We addressed this by storing manifests in version control as the canonical extraction record.

4.3 Entity Resolution (the “CFR is not ECFR” problem)

String similarity alone is dangerous for deduplication. The sequence “CFR” vs “ECFR” scores 0.857 on difflib—above many reasonable thresholds—yet one is the US Council on Foreign Relations and the other is the European Council on Foreign Relations. Pure auto-merge at any threshold below 0.95 would incorrectly collapse them.

Our two-pass approach combines string similarity for safe cases (≥ 0.95 : typos, punctuation) with LLM judging for borderline pairs (0.30–0.95). The LLM receives both raw names *plus* up to three evidence phrases from each side with extracted role and reasoning. It correctly distinguishes CFR and ECFR based on the evidence context (US vs European foreign policy), while correctly merging “Radio Netherlands” with “Radio Netherlands Worldwide” (rename detected) and “FEPS” with “Foundation for European Progressive Studies” (acronym expansion). Every decision is logged with similarity scores, evidence excerpts, and reasoning.

4.4 Source Mismatch

The batch-loading pipeline initially attached a generic citation (“Wikipedia—Von der Leyen Commission II page”) to 595 facts, even when the actual evidence came from individual commissioner biographies. This meant that a fact about Jacques Barrot’s service on the Prodi Commission (1999) was cited to a page about the Von der Leyen Commission (2024). We addressed this by migrating provenance records to the most specific available source document for each fact.

5 Findings

The methodology surfaces several substantive findings:

5.1 Collapse of the French Elite Pipeline

French commissioners from Santer through Von der Leyen I were consistently products of the *grandes écoles*: Pascal Lamy (ENA + Sciences Po + HEC), Pierre Moscovici (ENA + Sciences Po), Jacques Barrot (Sciences Po). The current French commissioner, Stéphane Séjourné, attended the University of Poitiers—a mid-tier regional university with no elite pipeline connection. He is the first French commissioner since 1995 with no Paris education of any kind. Across all commissions, elite institution concentration has declined from 37% (Barroso I) to 8% (Von der Leyen II).

5.2 Elite Network Asymmetry

Commissioners share 12 transnational elite networks with multiple members each (Bilderberg Group: 10, ECFR: 9, Trilateral Commission: 7). MEP leaders, across 136 identified SDT-relevant individuals, have *zero* exclusive elite networks with multiple members. The only networks crossing both groups are Friends of Europe (12 commissioners, 3 MEPs), Trilateral Commission (7/2), and the World Economic Forum (4/1). Commissioners operate in institutionalised, overlapping elite circuits; MEP leaders maintain fragmented, personal, non-overlapping networks.

5.3 EP-to-Commission Circulation

Only six of 150 commissioners held EP leadership positions (President, Vice President, or committee chair) before their Commission appointment. Circulation peaked at two per term under Barroso and Juncker, and has fallen to zero in the current Von der Leyen II term. When

commissioners do have MEP backgrounds, they are typically national ministers who served brief MEP stints as a transit lounge between national power and Brussels.

6 Conclusion

This paper’s own bibliography was verified using the methodology it describes. Each of the six cited works was downloaded as a PDF, checked by `check_citations.py`—a script that extracts citation claims from the LaTeX source, searches the cited PDF text, and sends numbered phrases to an LLM for phrase-level verification with confidence scoring. At the time of writing, all six citations pass at confidence ≥ 3 on the 1–5 scale described in the System Architecture. The verification report is committed alongside this paper and can be regenerated by any reader with API access.

The key methodological claim of this paper is that LLM-assisted research can be rigorous if the LLM is treated as a hypothesis generator rather than an authority. The specific mechanisms—phrase-indexed manifests, mandatory provenance, cross-validation against source text, and contextualised LLM judging for entity resolution—are transferable to other domains. The case study demonstrates that this discipline yields not just auditable data but substantive findings that manual methods would require years to surface.

References

- [1] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Yejin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38, 2022.
- [2] Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam Bowman, Stanislav Fort, Deep Ganguli, Danny Hernandez, Josh Jacobson, Jackson Kernion, Shauna Kravec, Liane Lovitt, Kamal Ndousse, Catherine Olsson, Sam Ringer, Dario Amodei, Tom Brown, Jack Clark, Nicholas Joseph, Ben Mann, Sam McCandlish, Chris Olah, and Jared Kaplan. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.
- [3] Joshua D. Kertzer and Jonathan Renshon. Experiments and surveys on political elites. *Annual Review of Political Science*, 2022.
- [4] Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of EMNLP*, 2023.
- [5] C. Wright Mills. *The Power Elite*. Oxford University Press, 1956.
- [6] Jiajie Zhang, Yushi Bai, Xin Lv, Wanjuan Gu, Danqing Liu, Minhao Zou, Shulin Cao, Lei Hou, Yuxiao Dong, Ling Feng, and Juanzi Li. LongCite: Enabling LLMs to generate fine-grained citations in long-context QA. *arXiv preprint arXiv:2409.02897*, 2024.